How AI and automation are shaping the future of Fonterra's operations

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Innovation has long been part of our Co-op's DNA.

For over a century, we have been finding new ways to make the most of every drop of our farmers' milk. Today, that commitment to continuous improvement is being accelerated by Artificial Intelligence (AI) and automation, which are transforming how we operate across our sites and supply chain.

Smarter quality control – one bag at a time

AI is playing a crucial role in advancing quality control, process efficiency and packaging reliability. One standout example is on our 25kg milk powder bagging lines, where AI-enabled image recognition technology now checks every bag for damage on 56 packing lines. Developed in-

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house by our Automation & Operational Technology team, the system automatically rejects bags that don't meet quality standards and provides timestamped images for traceability – helping prevent issues before they escalate.

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Every one of the 66 million bags filled annually is now visually inspected by AI, reducing waste and downtime while maintaining product quality. Building on this success, the technology has been adapted at our Clandeboye site to monitor bulk butter packaging. Here, AI tracks multiple stages of the packing process and can pause production if a fault is detected, allowing operators to intervene early. These systems are improving accuracy, preventing faults and increasing overall reliability.

Automation in packing and palletising

Across many of our packing and palletising plants, automation is now the norm. These facilities run 24/7 with minimal human intervention, supported by Automated Guided Vehicles (AGVs) – driverless forklifts designed specifically for our Co-op's operations. AGVs handle tasks such as loading empty bag pallets, unloading finished pallets and restacking them at designated drop-off points, ensuring smooth and efficient operations.

Most of our packing process is highly automated. Bags are de-stacked, opened, filled and heat-sealed without manual input. Once sealed, they pass through our in-house image cognition system, which uses cameras and machine learning to inspect every bag for quality issues. After checking the weight, foreign matter detection, and printing of traceable information on the bags, the bags are palletised by robots, stretch wrapped, labelled and transported by AGVs.

To further enhance quality assurance, we're working to replace manual Product Customer Acceptance (PCA) checks with advanced computer vision tools such as LiDAR, a laser-based technology that helps detect shapes, measure distances, and spot defects, and Optical Character Recognition (OCR), a system of verifying packaging information. These technologies will assess pallet stability, ensure correct labelling, and verify that products are securely packed – bringing greater consistency and coverage to our quality checks.

From predictive asset maintenance to optimising production workflows, we're actively exploring how this technology can drive greater value and efficiency across the entire supply chain.

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Smarter scheduling

Efficiency isn't just about machines – it's also about how we plan and coordinate. That's where

our Plant Equipment Scheduling Programme (ESP) comes in. The new programme replaces legacy spreadsheets with a centralised tool, improving visibility of asset availability and enabling better use of time. Whether it's scheduling maintenance, coordinating shared resources or planning production, Plant ESP supports informed and timely decisions.

Plant ESP leverages advanced AI-driven optimisation models and data products to enhance scheduling accuracy and operational efficiency. It integrates live data and site constraints to generate optimal schedules, balancing processing capacity, energy efficiency, and liquids buffer management (e.g. carryovers). Future enhancements include network-wide optimisation, workforce-aware scheduling, and real-time scenario modelling through AI agents and automation.

After a successful pilot at Edgecumbe, Clandeboye is now rolling out with targeted training support. Sites are already seeing improved coordination and fewer disruptions due to real-time data access. The next phase will focus on multi-site optimisation, the integration of workforce considerations, and enhanced responsiveness through AI-driven scenario modelling and decision support.

Early feedback has been positive, and engagement is strong across operational teams. Ongoing refinement and collaboration will underpin a resilient and efficient season.

Expanding AI across the supply chain

Beyond packaging and scheduling, AI is becoming increasingly embedded across the Co-op's broader operations. From predictive asset maintenance to optimising production workflows, we're actively exploring how this technology can drive greater value and efficiency across the entire supply chain.

Our Engineering teams are applying AI and large-scale industrial data operations within our Asset Performance Management platform. By analysing data from multiple sources and integrating predictive models, we're making faster, data-driven decisions. This allows us to apply operational context to maintenance strategies, improving how we allocate resources across our asset base.

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What's next: Autonomous Mobile Robots?

Looking ahead, the Co-op is exploring the use of Autonomous Mobile Robots (AMRs). Unlike AGVs, AMRs use cameras and sensors to navigate factory spaces more freely. They detect obstacles, avoid collisions and safely carry out repetitive transport tasks. This next level of automation promises enhanced safety and operational efficiency as we continue to evolve our operations.

As AI and automation continue to evolve, we are well-positioned to harness these technologies to drive smarter, safer and more efficient operations across our Co-op. From the factory floor to the supply chain, these innovations are helping us deliver better outcomes for our customers, farmer owners, and employees – ensuring we can stay ahead in a rapidly changing world.

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